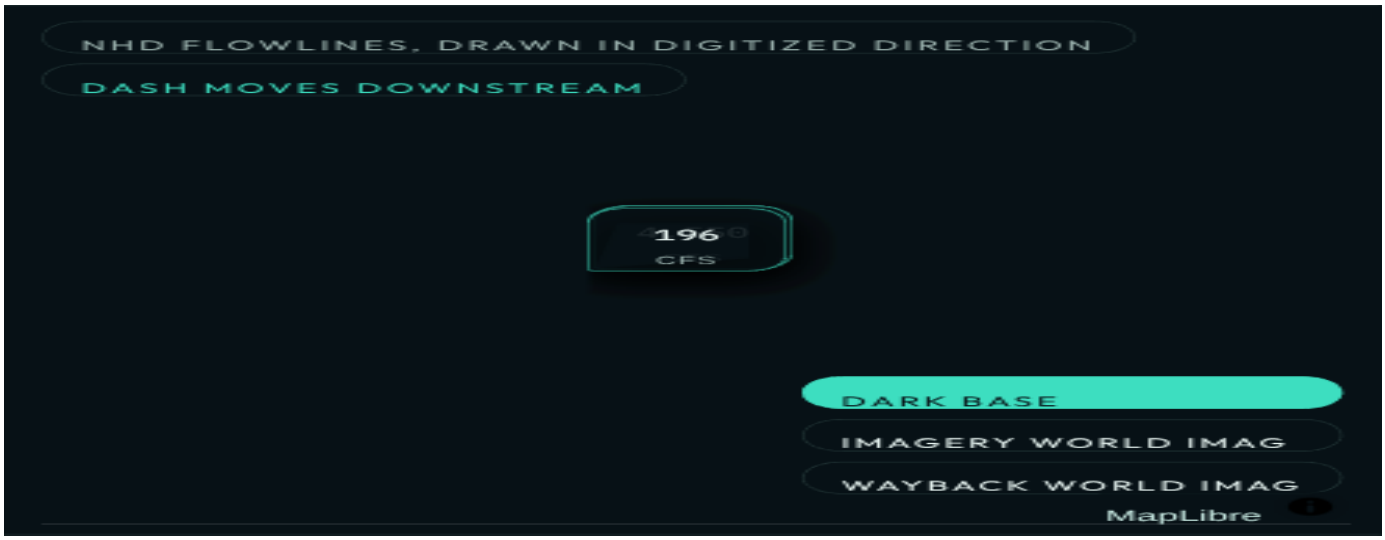
HydroTrust AI

Hydrological screening brief · IEEE ClimateChain 2026-10-17 15:28:45.802 UTC

Iowa City, Iowa, United States

41.66110, -91.53020 · stream source USGS NHD



Flood vulnerability

Index 53/100 · confidence high. Proximity 3.7/40, slope 25.9/30, discharge 23/30.

Elevation 203.1 m · slope 2.46% · nearest stream 829 m.

Nearest mapped stream is 829 m from the point (proximity 3.7/40).

The point sits 2.0 m below the surrounding 250 m samples, which adds ponding potential.

Cross-slope is 2.46% at 203.1 m elevation (slope 25.9/30).

Nearest discharge reading is 3,950 ft³/s at Iowa River at Iowa City, IA (05454500), 1,023 m away (flow 23.0/30).

Flood vulnerability index is 53/100. This is a screening score, not a regulatory floodplain determination.

Blue carbon

Rating 100/100 · 64.34 ha · 244.59 tCO2e/yr · dominant type emergent marsh.

54 wetland polygons cover 64.34 ha.

Dominant class is emergent marsh at 7.2 tCO,e/ha/yr.

Estimated sequestration is 244.59 tCO,e/yr, which rates 100/100 on a saturating local

Gauges

05454500 Iowa River at Iowa City, IA · 3,950 cfs · 14.15 ft · 1,023 m

05455010 South Branch Ralston Creek at Iowa City, IA · 0 cfs · 1.83 ft · 2,160 m

05454090 Muddy Creek at Coralville, IA · 8 cfs · 2.33 ft · 5,102 m

05454000 Rapid Creek near Iowa City, IA · 40 cfs · 4.26 ft · 5,586 m

05454300 Clear Creek near Coralville, IA · 111 cfs · 1.83 ft · 5,954 m

05453520 Iowa River below Coralville Dam nr Coralville, IA · 4,550 cfs · 52.55 ft · 6,026 m

05455100 Old Mans Creek near Iowa City, IA · 196 cfs · 3.45 ft · 9,354 m

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Advisory

Iowa City screens as elevated flood exposure (53/100).

The nearest live gauge, Iowa River at Iowa City, IA (05454500), is reading 3,950 ft³/s at a stage of 14.15 ft. Mapped wetlands cover 64.34 ha and are estimated to sequester 244.59 tCO,e/yr, led by en 3.7/40, slope 25.9/30, and discharge 23.0/30. Confidence is high because a live gauge, elevation samples, and surveyed streams all contributed. The index is a screening aid for community briefing, not a FEMA map.

Source: deterministic scorer.

No model key was configured, or the model call did not return valid JSON. Numbers above are computed locally from the telemetry snapshot.

Community actions

1. Walk the nearest low crossings and note which roads lose a shoulder first when stage rises.

2. Treat the mapped wetland edge as a buffer: avoid new fill, and photograph the vegetation line for the next seasonal comparison.

3. Check site 05454500 on a fixed cadence during wet weeks rather than only after a warning.

Policy actions

1. Use the index as a triage layer, then commission a site study before changing a land-use designation.

2. Count the estimated sequestration in any local stewardship credit, and keep the wetland polygon hash with the decision record.

3. Archive the source list with the score so a later reader can see whether NHD, OpenStreetMap, or both were live.

Verification

Content hash 0x5a07f64c6448e6d58eb0ba32d4b234fff8681c565b6420b4ace45fab90280997

IPFS CID bafkreia4qn7qhtbl3yin2wfnc2j55gehtyvb2bzjv6bjlmvzrkhd2ks5au

WEB3_STORAGE_TOKEN is not set. The CID is a local raw CIDv1 and has not been pinned.

470 dREC. Carbon term is floor(sequestration × 10) = 2445; on-chain cap is (100

No telemetry transaction has been confirmed for this briefing.

Sources

USFWS National Wetlands Inventory · USGS National Hydrography Dataset · OpenStreetMap Overpass · USGS Instantaneous Values · USGS Elevation Point Query Service · Esri imagery catalog

Screening product. Do not use as a flood insurance determination, engineering design, or carbon-registry issuance without a qualified review.